Solution: Subset Sum

Problem Statement

Given a set of positive numbers, determine if there exists a subset whose sum is equal to a given number 'S'.

Example 1:

```
Input: {1, 2, 3, 7}, S=6
Output: True
The given set has a subset whose sum is '6': {1, 2, 3}
```

Example 2:

```
Input: {1, 2, 7, 1, 5}, S=10
Output: True
The given set has a subset whose sum is '10': {1, 2, 7}
```

Example 3:

```
Input: {1, 3, 4, 8}, S=6
Output: False
The given set does not have any subset whose sum is equal to '6'.
```

Basic Solution

This problem follows the **0/1 Knapsack pattern** and is quite similar to `"Equal Subset Sum Partition"`. A basic brute-force solution could be to try all subsets of the given numbers to see if any set has a sum equal to 'S'.

So our brute-force algorithm will look like:

```
for each number 'i'
  create a new set which INCLUDES number 'i' if it does not exceed 'S', and recursively
     process the remaining numbers
  create a new set WITHOUT number 'i', and recursively process the remaining numbers
return true if any of the above two sets has a sum equal to 'S', otherwise return false
```

Since this problem is quite similar to `"Equal Subset Sum Partition"`, let's jump directly to the bottom-up dynamic programming solution.

Bottom-up Dynamic Programming

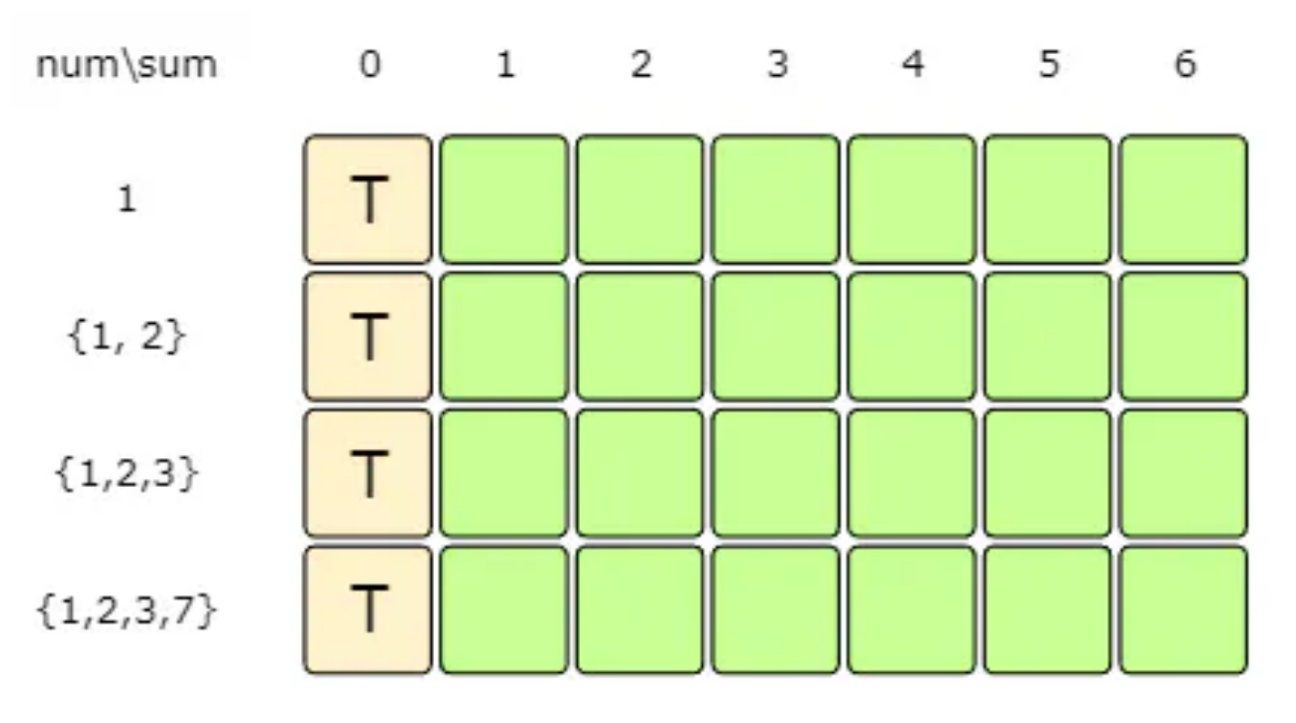
We'll try to find if we can make all possible sums with every subset to populate the array `dp[TotalNumbers][S+1]`.

For every possible sum ‘s’ (where 0 <= s <= S), we have two options:

1. Exclude the number. In this case, we will see if we can get the sum ‘s’ from the subset excluding this number => `dp[index-1][s]`
2. Include the number if its value is not more than ‘s’. In this case, we will see if we can find a subset to get the remaining sum => `dp[index-1][s-num[index]]`

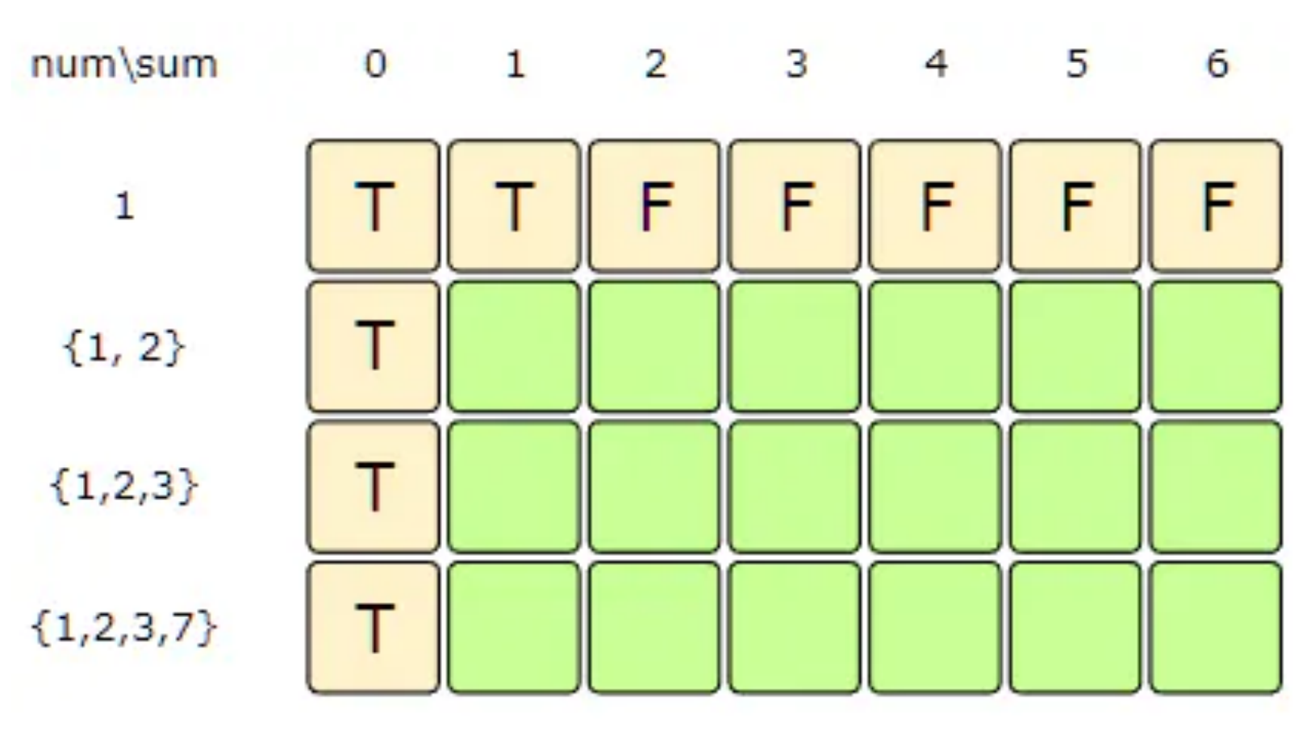
If either of the above two scenarios returns true, we can find a subset with a sum equal to ‘s’.

Let's draw this visually, with the example input {1, 2, 3, 7}, and start with our base case of size zero:



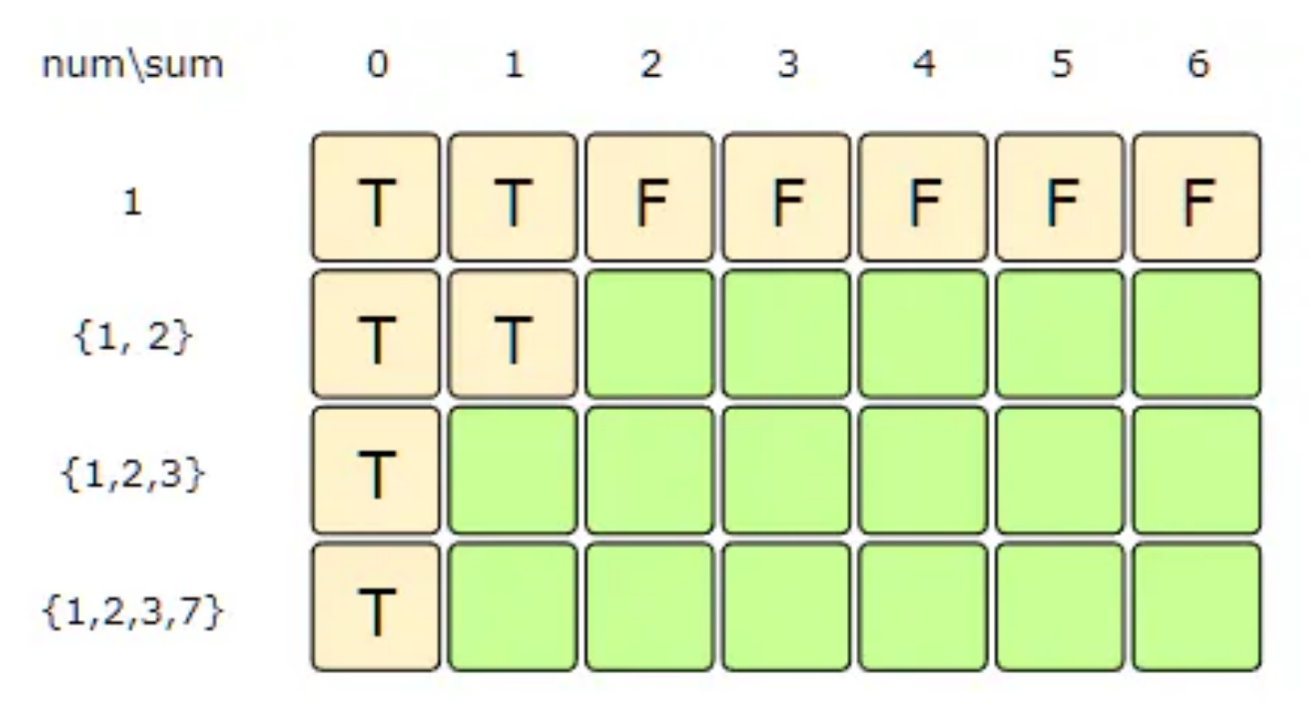
num\sum	0	1	2	3	4	5	6
1	T						
{1, 2}	T						
{1,2,3}	T						
{1,2,3,7}	T						

'0' sum can always be found through an empty set



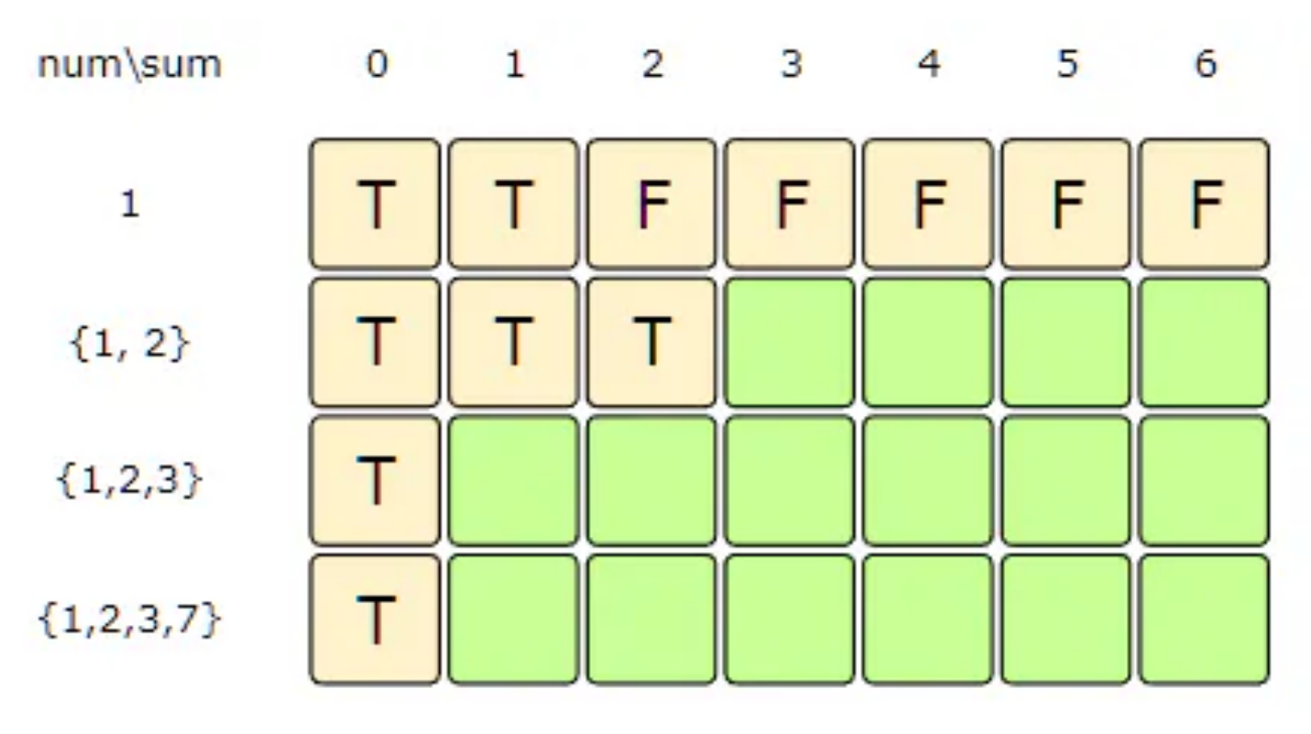
num\sum	0	1	2	3	4	5	6
1	T	T	F	F	F	F	F
{1, 2}	T						
{1,2,3}	T						
{1,2,3,7}	T						

With only one number, we can form a subset only when the required sum is equal to that number



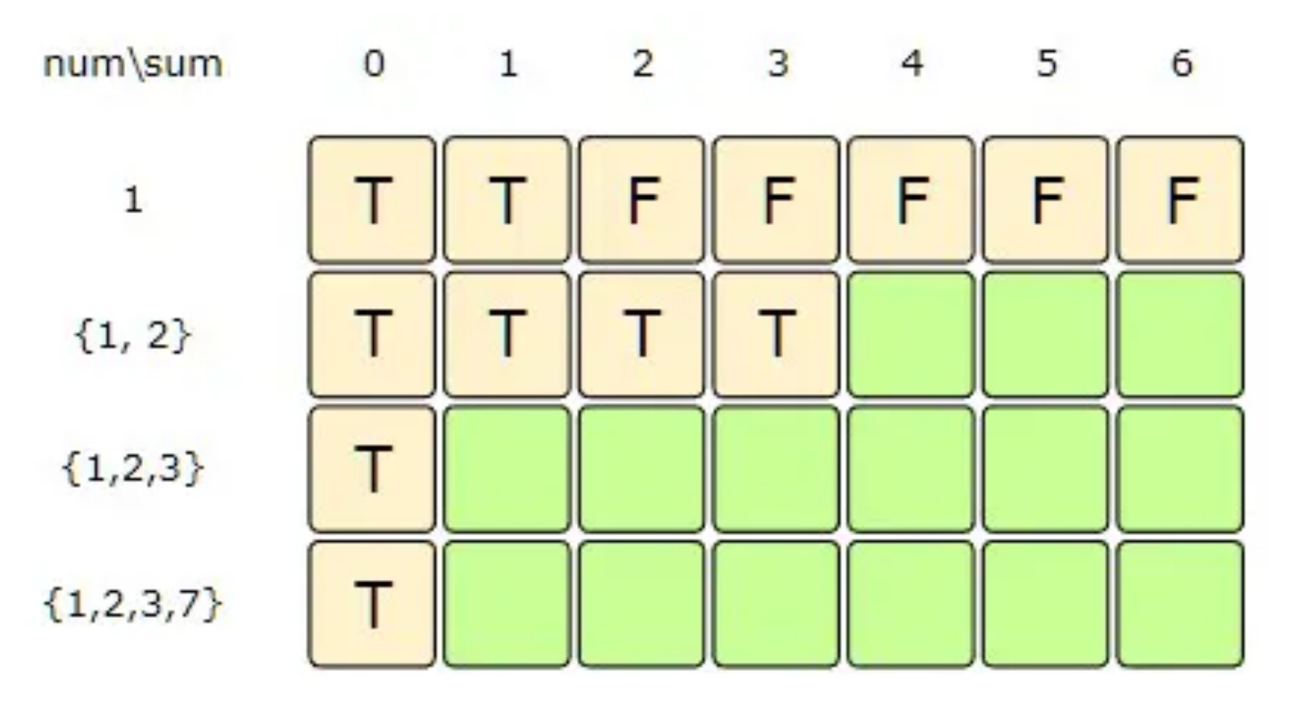
num\sum	0	1	2	3	4	5	6
1	T	T	F	F	F	F	F
{1, 2}	T	T					
{1,2,3}	T						
{1,2,3,7}	T						

sum: 1, index:1=> (dp[index-1][sum] , as the 'sum' is less than the number at index '1' (i.e., 1 < 2)



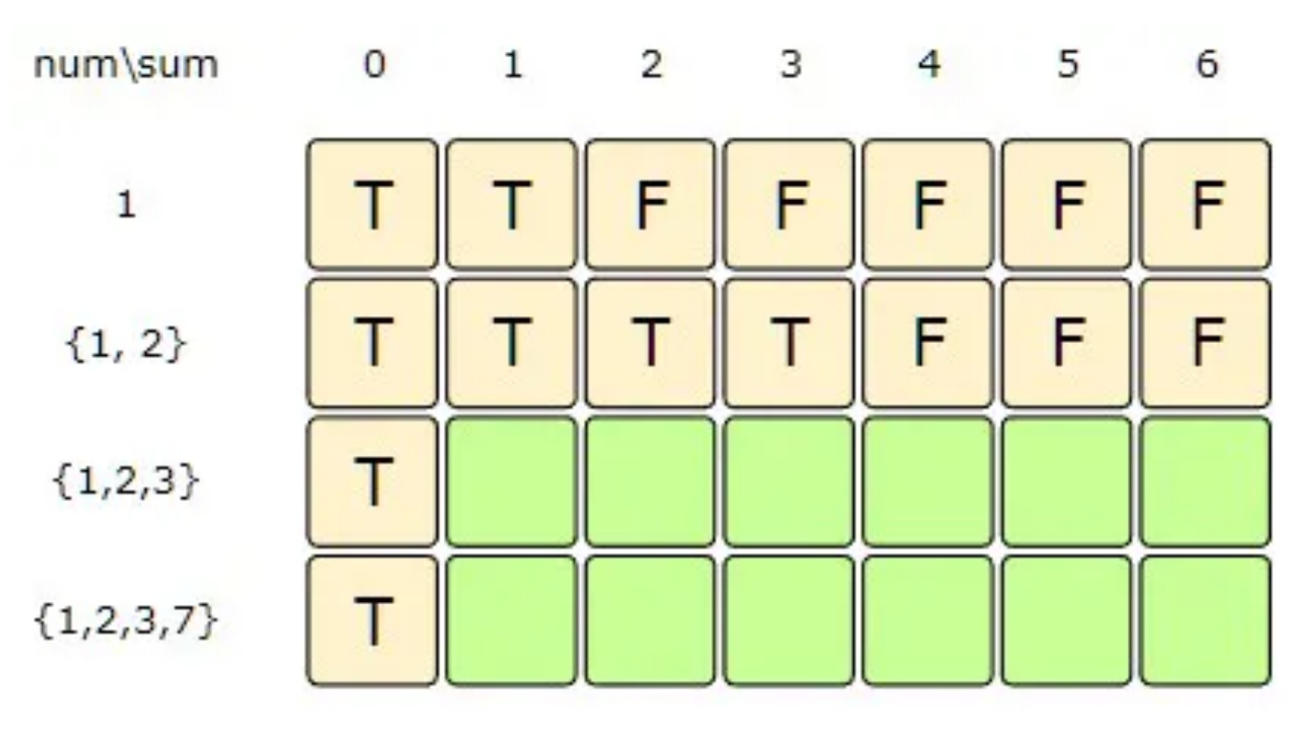
num\sum	0	1	2	3	4	5	6
1	T	T	F	F	F	F	F
{1, 2}	T	T	T				
{1,2,3}	T						
{1,2,3,7}	T						

sum: 2, index:1=> (dp[index-1][sum] || dp[index-1][sum-2])



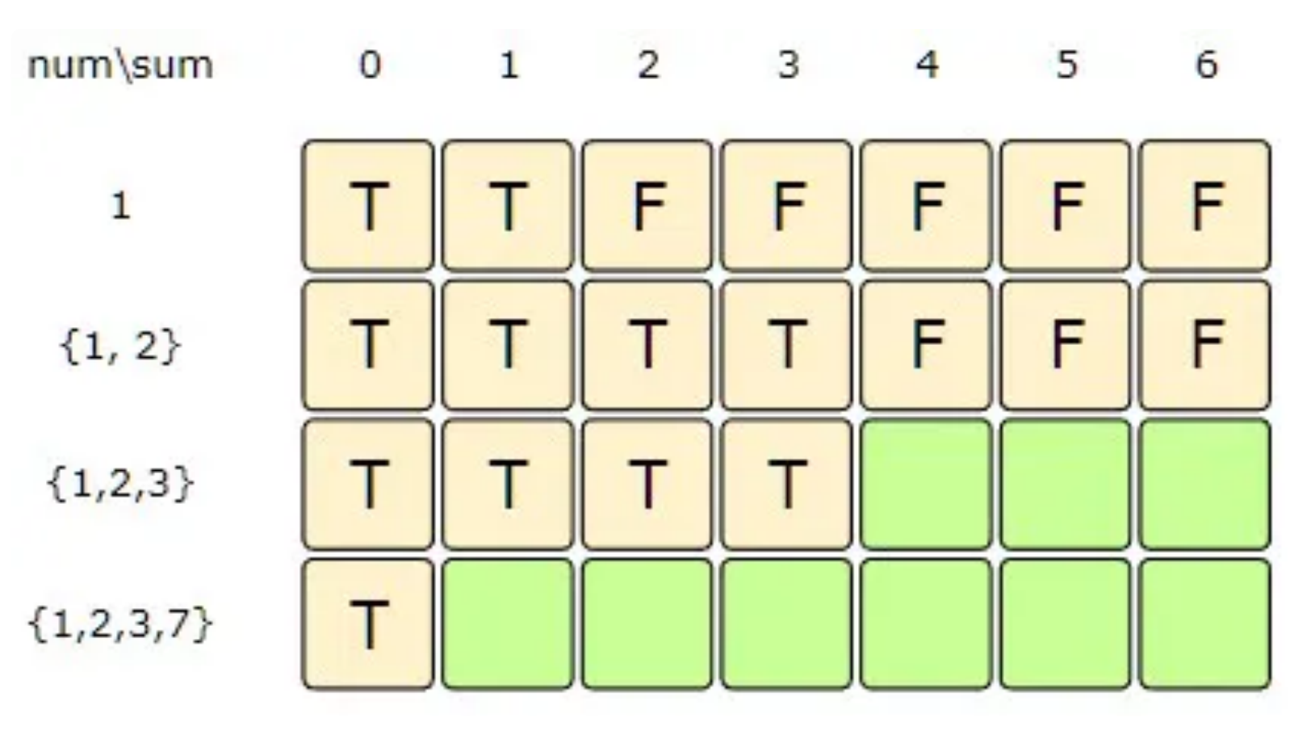
num\sum	0	1	2	3	4	5	6
1	T	T	F	F	F	F	F
{1, 2}	T	T	T	T			
{1,2,3}	T						
{1,2,3,7}	T						

sum: 3, index:1=> (dp[index-1][sum] || dp[index-1][sum-2])



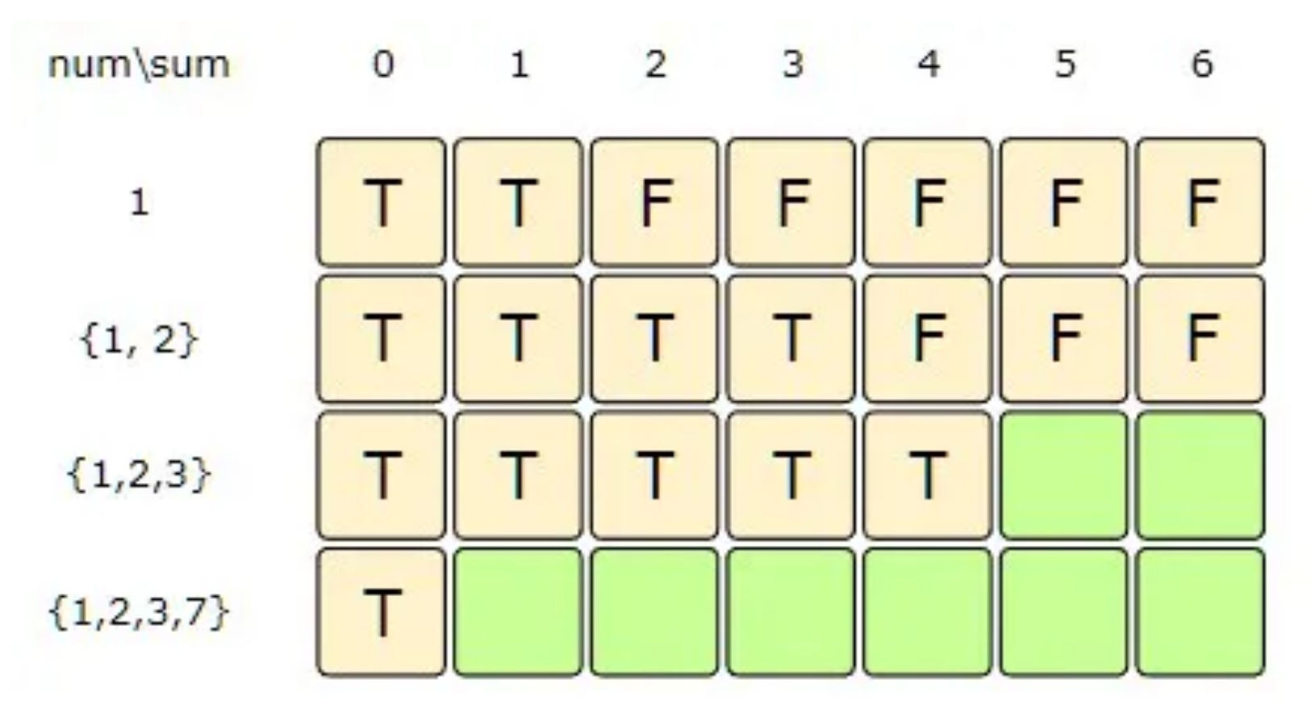
num\sum	0	1	2	3	4	5	6
1	T	T	F	F	F	F	F
{1, 2}	T	T	T	T	F	F	F
{1,2,3}	T						
{1,2,3,7}	T						

sum: 4-6 index:1=> (dp[index-1][sum] || dp[index-1][sum-2])



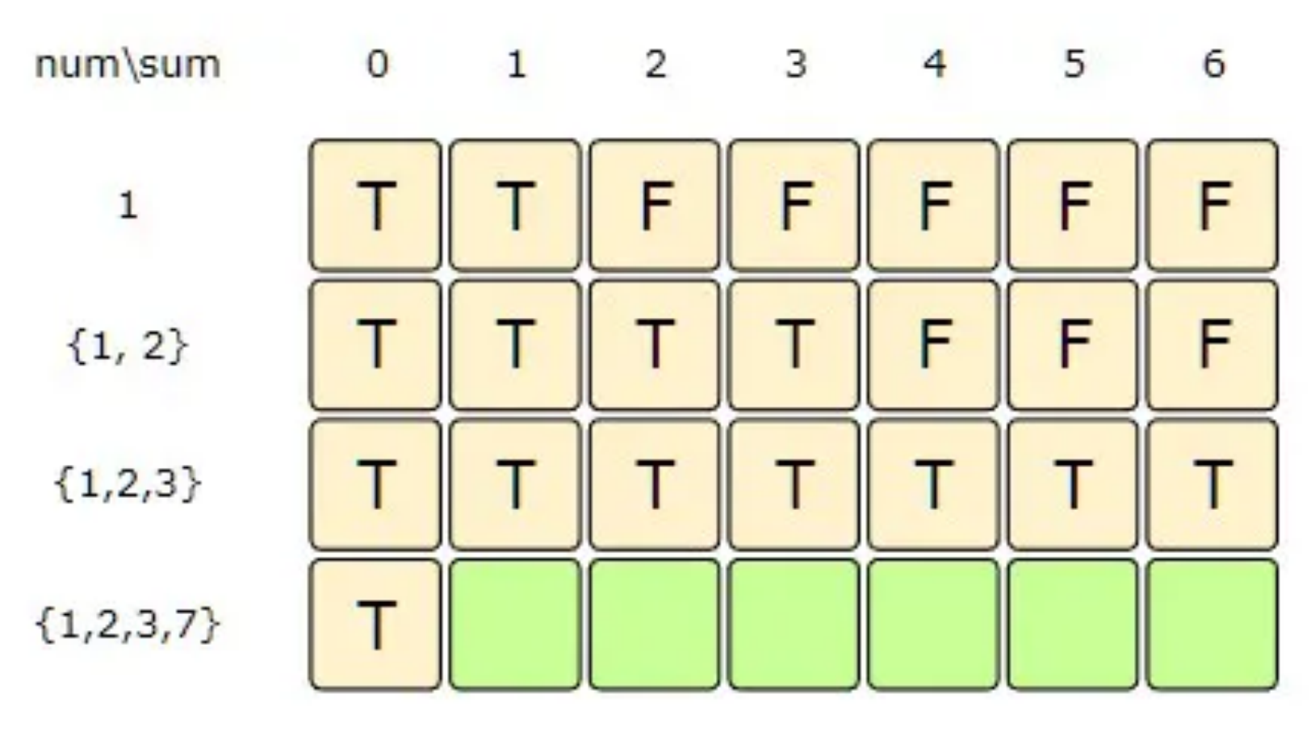
num\sum	0	1	2	3	4	5	6
1	T	T	F	F	F	F	F
{1, 2}	T	T	T	T	F	F	F
{1,2,3}	T	T	T	T			
{1,2,3,7}	T						

sum: 1,2,3, index:2=> (dp[index-1][sum] || dp[index-1][sum-3])



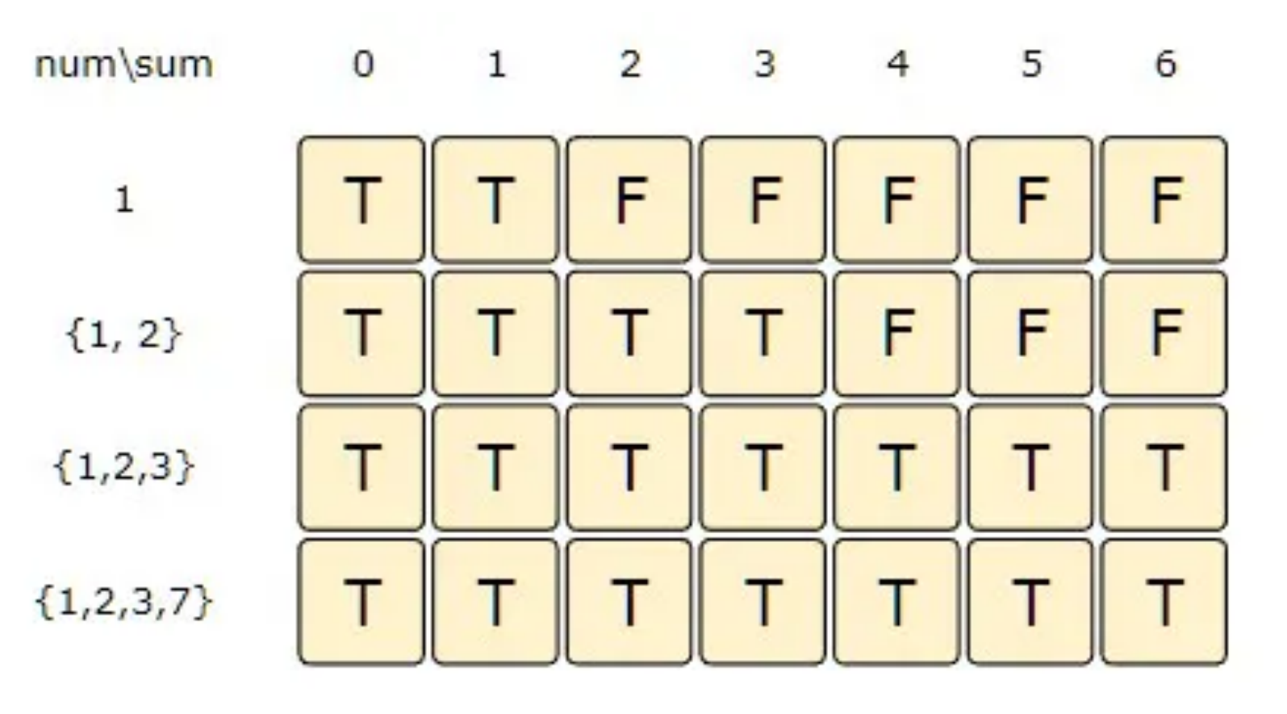
num\sum	0	1	2	3	4	5	6
1	T	T	F	F	F	F	F
{1, 2}	T	T	T	T	F	F	F
{1,2,3}	T	T	T	T	T		
{1,2,3,7}	T						

sum: 4, index:2=> (dp[index-1][sum] || dp[index-1][sum-3])



num\sum	0	1	2	3	4	5	6
1	T	T	F	F	F	F	F
{1, 2}	T	T	T	T	F	F	F
{1,2,3}	T	T	T	T	T	T	T
{1,2,3,7}	T						

sum: 5, 6, index:2=> (dp[index-1][sum] || dp[index-1][sum-3])

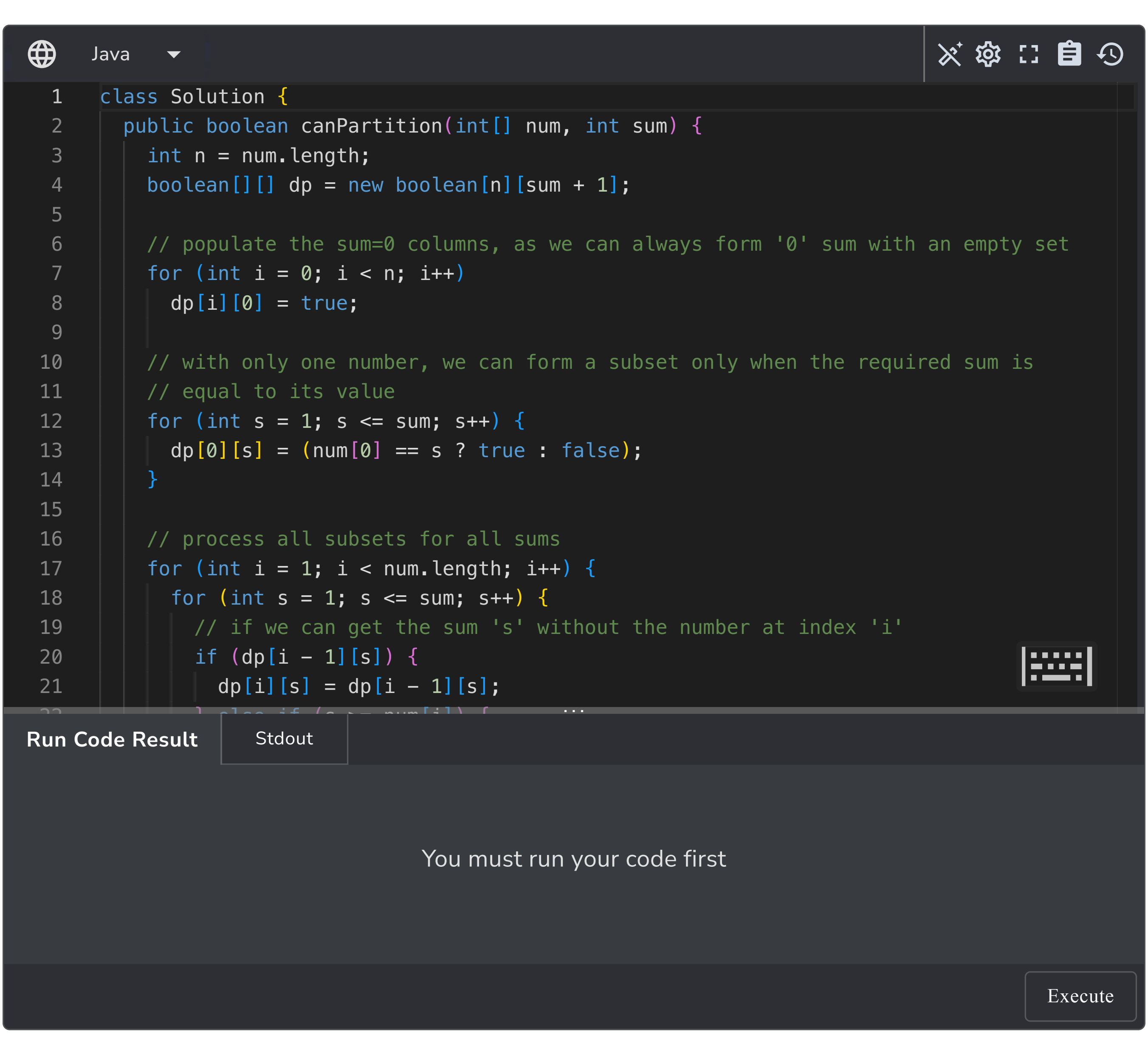


num\sum	0	1	2	3	4	5	6
1	T	T	F	F	F	F	F
{1, 2}	T	T	T	T	F	F	F
{1,2,3}	T	T	T	T	T	T	T
{1,2,3,7}	T	T	T	T	T	T	T

sum: 1-6, index:3=> (dp[index-1][sum] || dp[index-1][sum-7])

Code

Here is the code for our bottom-up dynamic programming approach:



```java
class Solution {
  public boolean canPartition(int[] num, int sum) {
    int n = num.length;
    boolean[][] dp = new boolean[n][sum + 1];

    // populate the sum=0 columns, as we can always form '0' sum with an empty set
    for (int i = 0; i < n; i++)
      dp[i][0] = true;

    // with only one number, we can form a subset only when the required sum is
    // equal to its value
    for (int s = 1; s <= sum; s++) {
      dp[0][s] = (num[0] == s ? true : false);
    }

    // process all subsets for all sums
    for (int i = 1; i < num.length; i++) {
      for (int s = 1; s <= sum; s++) {
        // if we can get the sum 's' without the number at index 'i'
        if (dp[i - 1][s]) {
          dp[i][s] = dp[i - 1][s];
```

The above solution has time and space complexity of $O(N * S)$, where ‘N’ represents total numbers and ‘S’ is the required sum.

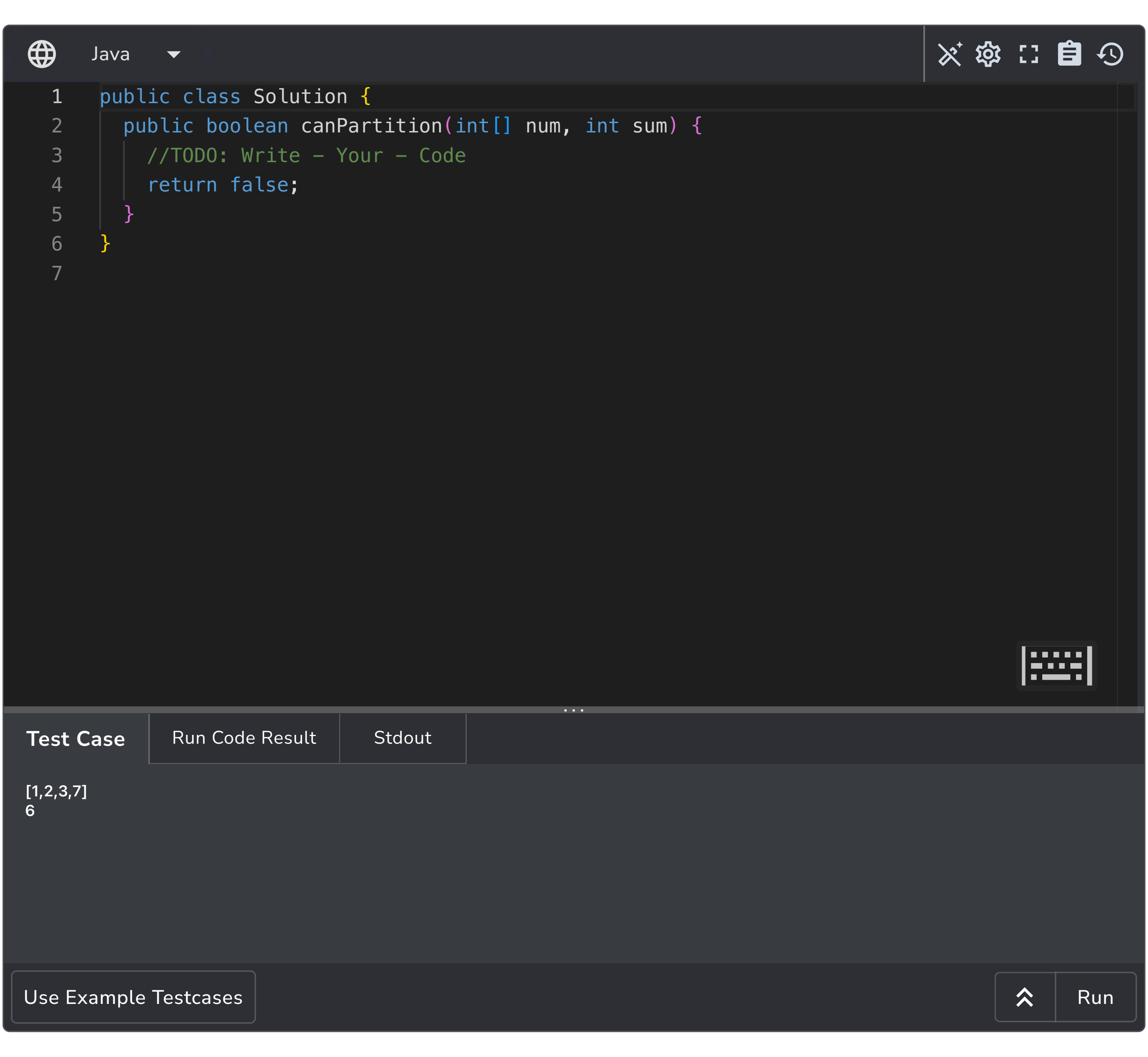
Challenge

Can we further improve our bottom-up DP solution? Can you find an algorithm that has $O(S)$ space complexity?

Hint

Similar to the space optimized solution for `"0/1 Knapsack"`.

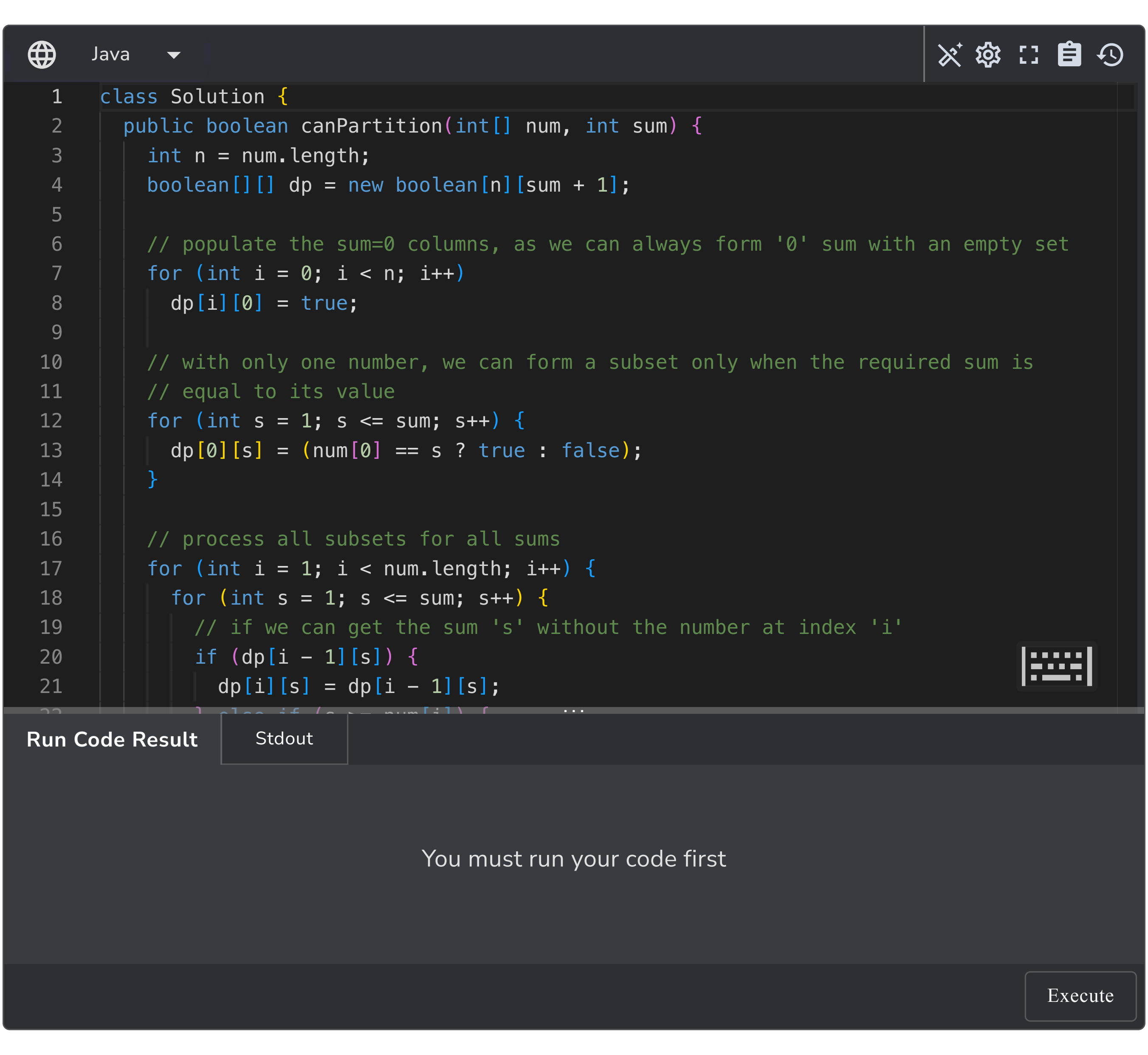
Try it yourself



```java
public class Solution {
  public boolean canPartition(int[] num, int sum) {
    //TODO: Write - Your - Code
    return false;
  }
}
```

```
[1,2,3,7]
6
```

Solution



```java
class Solution {
  public boolean canPartition(int[] num, int sum) {
    int n = num.length;
    boolean[][] dp = new boolean[n][sum + 1];

    // populate the sum=0 columns, as we can always form '0' sum with an empty set
    for (int i = 0; i < n; i++)
      dp[i][0] = true;

    // with only one number, we can form a subset only when the required sum is
    // equal to its value
    for (int s = 1; s <= sum; s++) {
      dp[0][s] = (num[0] == s ? true : false);
    }

    // process all subsets for all sums
    for (int i = 1; i < num.length; i++) {
      for (int s = 1; s <= sum; s++) {
        // if we can get the sum 's' without the number at index 'i'
        if (dp[i - 1][s]) {
          dp[i][s] = dp[i - 1][s];
```